

EE5203 L03 Starter Files

Download [EE5203_L03_starter.zip](#). It holds four files:

- Core/Inc/sensor_math.h
- Core/Inc/status_text.h
- Core/Src/sensor_math.c
- Core/Src/status_text.c

Add them to your project

1. Copy your EE5203_L02_<studentnumber> project, rename the copy EE5203_L03_<studentnumber>, and delete its build folder.
2. Double-click the zip. Copy the Core folder inside and paste it into your project folder. Windows adds the files to your Core\Inc and Core\Src.
3. In CMakeLists.txt, set the project name to the folder name and add the two .c files to target_sources:

```
set(CMAKE_PROJECT_NAME EE5203_L03_<studentnumber>)

target_sources(${CMAKE_PROJECT_NAME} PRIVATE
    # Add user sources here
    Core/Src/sensor_math.c
    Core/Src/status_text.c
)
```

Without step 3 the build stops with `undefined reference to 'clamp_u16'`. The L03 lab guide, Checkpoint A, has every step.

What the starter does

The functions compile, but they return placeholder results. Complete every TODO using the L03 lab guide. Add the arrays, the result variables, and the function calls to main.c only inside its USER CODE regions.

Before you copy the main.c declarations, be able to explain them:

- `uint16_t` is an unsigned 16-bit whole number: 0 to 65,535. It comes from `<stdint.h>`.
- Plain `int` is used for the small counts, indices, and capacities.
- `bool` stores `true` or `false`. It comes from `<stdbool.h>`.
- `#define SAMPLE_COUNT 8` is not a variable. The preprocessor replaces every `SAMPLE_COUNT` with `8` before the compiler runs.